Xiao ZHOU

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Education

The University of Oxford — Oxford, UK
D.Phil. in Engineering Science — *Oct. 2025 – Present*
Advisor: Prof. Nick Hawes, Oxford Robotics Institute (GOALS Group)

Topic: cooperation and coordination in multi-agent systems: MARL, graph neural networks, and combinatorial optimisation, with a focus on ad hoc teamwork.

The Hong Kong University of Science and Technology — Guangzhou, CHN
M.Phil. in Robotics and Autonomous Systems Thrust — *Sep. 2023 – Aug. 2025*
Advisor: Prof. Jun MA

Harbin Institute of Technology — Shenzhen, CHN
B.Eng. in Electronics and Information Engineering — *Sep. 2018 – Jun. 2023*

Selected Experience

Research Assistant — Mar. 2025 – Aug. 2025
Sycamore Lab — EPFL
Supervisor: Prof. Maryam Kamgarpour — Dept of Mechanical Engineering

- Developed a multi-agent reinforcement learning framework for fire rescue in dynamic hazardous environments.
- Proposed a safe MARL method that maximises reward under chance-constrained safety guarantees.

Research Assistant — Oct. 2022 – Jun. 2023
Robotics and Autonomous Systems Thrust — The Hong Kong University of Science and Technology
Supervisor: Prof. Jun MA — Dept of Electronic and Computer Engineering

- Modelled the interactions and intentions of vehicles in urban driving scenarios as a Markov game and proposed a general solution based on level-k game theory.
- Proposed a temporal-spatial attention-based deep Q-learning algorithm for decision-making in complex driving environments; published in *IEEE Trans. Intelligent Vehicles*.

Undergraduate Research Assistant — Jun. 2021 – Mar. 2022
AI & Robot Lab — Tsinghua University, Shenzhen
Supervisor: Prof. Xueqian WANG — School of Control Science and Engineering

- Deployed real-time SLAM and navigation on the AgileX SCOUT platform over extremely rough terrain.
- Implemented local planning with NMPC and global planning with RRT*.
- Built uncertainty-aware point-cloud maps of complex terrain using Gaussian process regression.

Review Activities

IEEE Robotics and Automation Letters (RA-L)

IEEE Transactions on Intelligent Vehicles (TIV)

IEEE Intelligent Transportation Systems Conference (ITSC)

IEEE International Conference on Intelligent Robots and Systems (IROS)

Publications & Preprints

- **Xiao Zhou***, Zengqi Peng, Yang Song, Junan Lin and Jun Ma, "SocialTraj: Two-Stage Socially-Aware Trajectory Prediction for Autonomous Driving via Conditional Diffusion Model," Accepted to *IEEE Transactions on Vehicular Technology*, 2026. [PDF]
- Benshan Ma, Haichao Liu, Ruiguo Zhong, Pei Liu, **Xiao Zhou**, Jun Ma, "Behavioral Uncertainty-Aware Attention Allocation via VLMs for Interactive Autonomous Driving," Accepted to *IEEE Transactions on Vehicular Technology*, 2026. [PDF]
- **Xiao Zhou***, Chengzhen Meng, Wenru Liu, Zengqi Peng, Ming Liu, and Jun Ma, "Integrated Intention Prediction and Decision-Making with Spectrum Attention Net and Proximal Policy Optimization," Accepted to *27th IEEE International Conference on Intelligent Transportation Systems (ITSC)*, 2024. Preprint available: `https://arxiv.org/abs/2408.03191`. [PDF] [Oral]
- Zengqi Peng, **Xiao Zhou**, Lei Zheng, Yubin Wang, Jun Ma, "Reward-Driven Automated Curriculum Learning for Interaction-Aware Self-Driving at Unsignalized Intersections," Accepted to *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2024. Preprint available: `https://arxiv.org/abs/2403.13674`. [PDF]
- **Xiao Zhou***, Zengqi Peng, Yusen Xie, Ming Liu, and Jun Ma, "Game-Theoretic Driver Modeling and Decision-Making for Autonomous Driving with Temporal-Spatial Attention-Based Deep Q-Learning," Accepted to *IEEE Transactions on Intelligent Vehicles*, 2024. [PDF] [Page]
- Zengqi Peng, **Xiao Zhou**, Yubin Wang, Lei Zheng, Ming Liu, and Jun Ma, "Curriculum Proximal Policy Optimization with Stage-Decaying Clipping for Self-Driving at Unsignalized Intersections," Accepted to *26th IEEE International Conference on Intelligent Transportation Systems (ITSC)*, 2023. Preprint available: `https://arxiv.org/abs/2308.16445`. [PDF]
- Zhuozhu Jian, Zihong Lu, **Xiao Zhou**, Bin Lan, Anxing Xiao, Xueqian Wang, Bin Liang, "PUTN: A Plane-fitting based Uneven Terrain Navigation Framework," Accepted to *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2022. Preprint available: `https://arxiv.org/abs/2203.04541`. [PDF] [Video]

Skills

Programming: Python, C/C++

ML / RL: PyTorch, JAX, Stable-Baselines3, TorchRL, JaxMARL

Infrastructure: Linux, Git, distributed training on HPC clusters (SLURM), Docker

Robotics & Tools: ROS, Gazebo, MATLAB/Simulink, Solidworks, LaTeX

Hardware: Arduino, Raspberry Pi, FPGA, motors and sensors, basic mechanical design

Honors & Awards

NCCR Automation Fellowship	2024
HKUST Redbird Postgraduate Scholarship	2023–2025
Excellent Graduation Thesis for Undergraduate Students	2023
Outstanding Students 2020–2021, Harbin Institute of Technology, Shenzhen	2021
First Prize, China Undergraduate Mathematical Contest in Modeling	2020